

Sensory Transduction in the Eye

The transduction of visual information to the nervous system begins with the light-induced conversion of *cis*-retinal to *trans*-retinal in rods and cones. Like other *cis-trans* pairs, these isomers of retinal differ only in the spatial arrangement of atoms at a carbon-carbon double bond (see Figure 4.7).

As shown in Figure 50.17, *trans*-retinal and *cis*-retinal differ in shape. This shift in shape activates the visual pigment (in rods, rhodopsin), which activates a G protein, which in turn activates the enzyme phosphodiesterase. The substrate for this enzyme in rods and cones is cyclic GMP (cGMP), which in the dark binds to sodium ion (Na^+) channels and keeps them open (Figure 50.18a). When the enzyme hydrolyzes cGMP, Na^+ channels close, and the cell becomes hyperpolarized (Figure 50.18b). The signal transduction pathway then shuts off as enzymes convert retinal back to the *cis* form, inactivating the visual pigment.

In bright light, rhodopsin remains active, and the response in the rods becomes saturated. If the amount of light entering the eyes abruptly decreases, the rods do not regain full responsiveness for several minutes. This is why you are briefly blinded if you pass quickly from bright sunshine into a dark movie theater. (Because light activation changes the color of rhodopsin from purple to yellow, rods in which the light response is saturated are often described as “bleached.”)

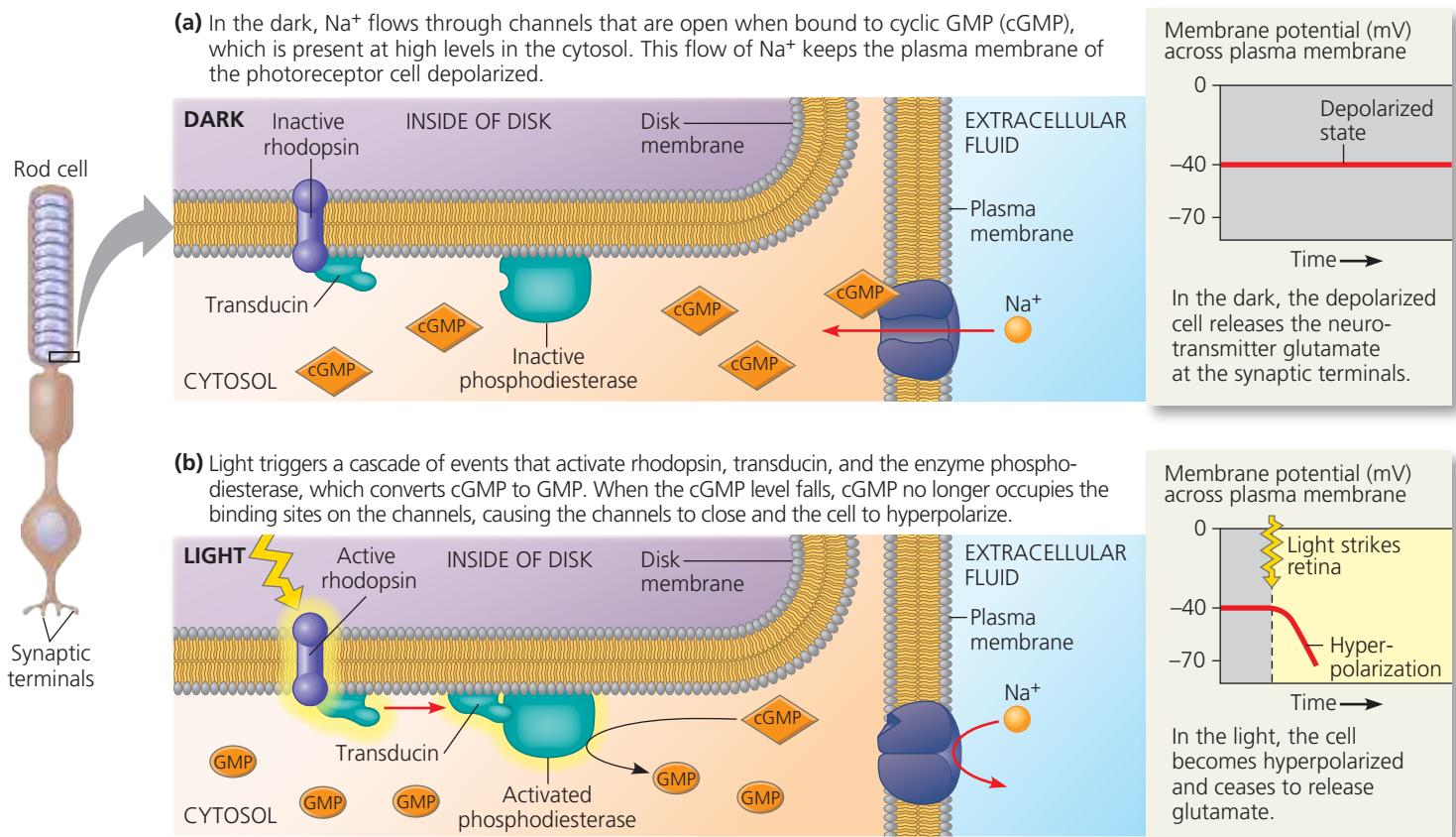
Processing of Visual Information in the Retina

The processing of visual information begins in the retina itself, where both rods and cones form synapses with bipolar cells (see Figure 50.17). In the dark, rods and cones are depolarized and continually release the neurotransmitter glutamate at these synapses (Figure 50.19). When light strikes the rods and cones, they hyperpolarize, shutting off their release of glutamate. This decrease triggers a change in the membrane potential of the bipolar cells, altering their regulation of action potential transmission to the brain.

Processing of signals from rods and cones occurs via several different pathways in the retina. Some information passes directly from photoreceptors to bipolar cells to ganglion cells. In other cases, horizontal cells carry signals from one rod or cone to other photoreceptors and to several bipolar cells.

How is it adaptive for visual information to follow several paths? We'll consider one example. When an illuminated rod or cone stimulates a horizontal cell, the horizontal cell inhibits more distant photoreceptors and bipolar cells that are not illuminated. The result is that the region receiving light appears lighter and the dark surroundings even darker. This form of integration, called *lateral inhibition*, sharpens edges and enhances contrast in the image. An essential part of visual processing, lateral inhibition occurs in the brain as well as the retina.

▼ **Figure 50.18 Response of a photoreceptor cell to light.** Light triggers a receptor potential in a rod (shown here) or cone. Note that for photoreceptors, this change in membrane potential is a hyperpolarization.



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